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Use of Relationship Between Activities of Different Traffic Signals in a Network to Improve Traffic Signal State Estimation

Abstract

Methods and devices for using a relationship between activities of different traffic signals in a network to improve traffic signal state estimation are disclosed. An example method includes determining that a vehicle is approaching an upcoming traffic signal. The method may further include determining a state of one or more traffic signals other than the upcoming traffic signal. Additionally, the method may also include determining an estimate of a state of the upcoming traffic signal based on a relationship between the state of the one or more traffic signals other than the upcoming traffic signal and the state of the upcoming traffic signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/406,797, filed on Jan. 8, 2024, which is a continuation of U.S. patent application Ser. No. 18/150,304 (now U.S. Pat. No. 11,900,693), filed on Jan. 5, 2023, which is a continuation of U.S. patent application Ser. No. 17/154,976 (now U.S. Pat. No. 11,568,652), filed on Jan. 21, 2021, which is a continuation of U.S. patent application Ser. No. 16/257,735 (now U.S. Pat. No. 10,909,396) filed on Jan. 25, 2019, which is a continuation of U.S. patent application Ser. No. 15/839,124 (now U.S. Pat. No. 10,210,408) filed on Dec. 12, 2017, which is a continuation of U.S. patent application Ser. No. 15/653,716 (now U.S. Pat. No. 9,875,417) filed on Jul. 19, 2017, which is a continuation of U.S. patent application Ser. No. 14/843,705 (now U.S. Pat. No. 9,734,417) filed on Sep. 2, 2015, which is a continuation of U.S. patent application Ser. No. 13/622,418 (now U.S. Pat. No. 9,158,980) filed on Sep. 19, 2012, the entire contents of each of which are hereby incorporated by reference.

BACKGROUND

[0002] Some vehicles are configured to operate in an autonomous mode in which the vehicle navigates through an environment with little or no input from a driver. Such a vehicle may include one or more sensors that are configured to sense information about the environment. The vehicle may use the sensed information to navigate through the environment.

[0003] For example, if an output of the sensors is indicative that the vehicle is approaching an obstacle, the vehicle may navigate around the obstacle. Additionally, a vehicle may sense information about traffic signs and traffic signals. For example, traffic signs may provide regulatory information or warning information while traffic signals positioned at road intersections, pedestrian crossings, and other locations may be used to control competing flows of traffic.

SUMMARY

[0004] In one example aspect, a method is disclosed that includes determining, using a computing device, that a vehicle is approaching an upcoming traffic signal. The method may further include determining, using the computing device, a state of one or more traffic signals other than the upcoming traffic signal. Additionally, the method may also include determining, using the computing device, an estimate of a state of the upcoming traffic signal based on a relationship

between the state of the one or more traffic signals other than the upcoming traffic signal and the state of the upcoming traffic signal.

[0005] In another example aspect, a non-transitory computer-readable medium is disclosed having stored therein instructions executable by a computing device to cause the computing device to perform functions. The functions may include determining that a vehicle is approaching an upcoming traffic signal. The functions may further include determining a state of one or more traffic signals other than the upcoming traffic signal. Additionally, the functions may also include determining an estimate of a state of the upcoming traffic signal based on a relationship between the state of the one or more traffic signals other than the upcoming traffic signal and the state of the upcoming traffic signal.

[0006] In yet another example aspect, a controller is disclosed. The controller may include at least one processor, a memory, and instructions stored in the memory and executable by the at least one processor to cause the controller to perform functions. The functions may include determining that a vehicle is approaching an upcoming traffic signal. The functions may further include determining a state of one or more traffic signals other than the upcoming traffic. Additionally, the functions may also include determining an estimate of a state of the upcoming traffic signal based on a relationship between the state of the one or more traffic signals other than the upcoming traffic signal and the state of the upcoming traffic signal.

[0007] The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the figures and the following detailed description.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0008] FIG. 1 is a block diagram of an example method of estimating a state of a traffic signal.

[0009] FIGS. 2A-2C are example conceptual illustrations of determining a state of a traffic signal.

[0010] FIG. 3 is an example flow chart for controlling a vehicle.

[0011] FIG. 4 is an example conceptual illustration of determining states of traffic signals in a road network to estimate a state of an upcoming traffic signal.

[0012] FIG. 5 illustrates an example vehicle, in accordance with an embodiment.

[0013] FIG. 6 is a simplified block diagram of an example vehicle, in accordance with an embodiment.

[0014] FIG. 7 is a schematic illustrating a conceptual partial view of an example computer program product that includes a computer program for executing a computer process on a computing device, arranged according to at least some embodiments presented herein.

DETAILED DESCRIPTION

[0015] In the following detailed description, reference is made to the accompanying figures, which form a part hereof. In the figures, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, figures, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure as generally describe herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

[0016] A vehicle, such as a vehicle configured to operate autonomously, may be configured to determine a state of a traffic signal. For example, the vehicle may use one or more sensors such as a camera laser scanner and/or LIDAR to determine a state of a traffic signal. In some instances,

estimating the state of an upcoming traffic signal (e.g., red, yellow, or green) may be challenging for a controller of a vehicle. Detection of a color of a traffic signal may be difficult because the color of the traffic signal may blend in with a background, a traffic signal may be swaying in the wind, or a traffic signal may have been temporarily relocated due to construction, for example. Therefore, incorporating as much additional information as possible about an estimate of the state of a traffic signal may be important to improve confidence in the estimate.

[0017] According to the described systems and methods, an estimate of a state of an upcoming traffic signal may be determined based on a relationship between the state of the upcoming traffic signal and states of one or more other traffic signals in the environment. For example, if a vehicle receives information about states of other traffic signals (e.g., traffic signals on side streets, traffic signals in the distance, or cross-traffic traffic signals) recently observed by the vehicle and/or other vehicles, a controller of the vehicle may use a known relationship between the states of the traffic signals and the upcoming traffic signal to provide an additional inference about the state of the upcoming traffic signal.

[0018] Although the vehicle is described as being configured to operate autonomously, the example is not meant to be limiting. In another instance, a computing device (e.g., a computing device that is on-board the vehicle or a computing device in a server) may be configured to estimate the state of the upcoming traffic signal and provide the estimate of the state of the upcoming traffic signal to another device, such as a driver assistance system or a navigation system. Subsequently, the estimate of the state of the upcoming traffic signal may be used by the another device to perform one or more functions. Additionally, although many of the examples are described with respect to an upcoming traffic signal for a vehicle, the described systems and methods may be applicable to estimating the state of any traffic signal in a road network. Various other example scenarios are also described hereinafter.

[0019] FIG. 1 is a block diagram of an example method **100** of estimating a state of a traffic signal. Method **100** shown in FIG. 1 presents an embodiment of a method that could be used with the vehicles described herein, for example, and may be performed by a vehicle or components of a vehicle, or more generally by a server or other computing device. Method **100** may include one or more operations, functions, or actions as illustrated by one or more of blocks **102-108**. Although the blocks are illustrated in a sequential order, these blocks may also be performed in parallel, and/or in a different order than those described herein. Also, the various blocks may be combined into fewer blocks, divided into additional blocks, and/or removed based upon the desired implementation.

[0020] In addition, for the method **100** and other processes and methods disclosed herein, the block diagram shows functionality and operation of one possible implementation of present embodiments. In this regard, each block may represent a module, a segment, or a portion of program code, which includes one or more instructions executable by a processor for implementing specific logical functions or steps in the process. The program code may be stored on any type of computer-readable medium, such as, for example, a storage device including a disk or hard drive. The computer-readable medium may include a non-transitory computer-readable medium, for example, such as computer-readable media that store data for short periods of time like register memory, processor cache, and Random Access Memory (RAM). The computer-readable medium may also include non-transitory media, such as secondary or persistent long term storage, like read only memory (ROM), optical or magnetic disks, and compact-disc read only memory (CD-ROM), for example. The computer-readable media may also be any other volatile or non-volatile storage systems. The computer-readable medium may be considered a computer-readable storage medium, a tangible storage device, or other article of manufacture, for example.

[0021] In addition, for the method **100** and other processes and methods disclosed herein, each block may represent circuitry that is configured to perform the specific logical functions in the process.

[0022] As shown, initially, at block **102**, the method **100** includes determining, using a computing device, that a vehicle is approaching an upcoming traffic signal. In one example, the computing device may be configured to control the vehicle in an autonomous mode. The vehicle described with respect to FIGS. **5** and **6** is one such example of a vehicle that may be configured to operate autonomously. The vehicle may include a GPS receiver (or other geographic positioning component) to determine a geographic location and an accelerometer, gyroscope, or other acceleration device to determine a pitch, yaw, and roll (or changes thereto) relative to the direction of gravity. In one instance, a geographic location of the vehicle may be compared to a map of an environment that includes information about intersections of roads to determine a proximity of the vehicle to an intersection known to include a traffic signal.

[0023] In a further example, if the vehicle is approaching a traffic signal as indicated on a map of positions of traffic signals, one or more sensors of the vehicle may be used to scan a target area to determine a state of the traffic signal. The one or more sensors may include imaging and/or non-imaging components. For example, various types of cameras may be mounted in various configurations to the vehicle to capture an image of an upcoming area. In one instance, a camera may be positioned to face straight ahead and mounted behind or near a rear-view mirror. Additionally, a camera may capture a specific region of interest, such as a 2040×1080 region, of a camera with a fixed lens with a 30 degree field of view. The camera may be calibrated to detect traffic signals at various distances to ensure a reasonable braking distance. In a further example, gain and shutter speeds of the camera may be set to avoid saturation of traffic signals during the day and/or night.

[0024] In another example, the one or more sensors may include a three-dimensional (3D) scanning device configured to determine distances to surfaces of objects in an upcoming area. In one instance, a structured light projection device and camera may be used to determine a three-dimensional point cloud describing the upcoming area. In another instance, a laser and/or radar device such as a LIDAR or laser rangefinder may scan the target area to determine distances to objects. In other instances, a stereo camera or time-of-flight camera may be used for range imaging.

[0025] Based on information from the one or more sensors, a traffic signal may be detected. In some instances, a pattern, template, shape, or signature that is expected for traffic signals may be identified. As an example, a traffic signal classifier may find a pattern of red, yellow, and green objects with appropriate size and aspect ratios within an image. Any example image processing techniques may be used to identify one or more portions of the image matching a known pattern. Template matching is one possible example.

[0026] In an instance, in which the 3D position of a traffic signal is known, a prediction of where a traffic signal should appear in an image may be used to determine the state of the traffic signal. For example, based on an estimate of a position of the vehicle with respect to the 3D position (e.g., using a GPS), a predicted position for the traffic signal can be projected into the image from of a camera of the vehicle. In one example, the predicted position may be an axis-aligned bounding box which selects a portion of the image. However, other example regions or shapes are also possible. The portion of the image may then be analyzed to determine the state of the traffic signal.

[0027] For example, the predicted position may be processed to identify brightly colored red or green objects, and the predicted position may change as the position of the vehicle approaches the traffic signal. In some instances, if a traffic signal is detected, a state of the traffic signal may be determined. Determining the state of the traffic signal may include determining which object in a pattern of red, yellow, and green objects of an image of a traffic signal is illuminated. In one instance, an image processing method may be used to determine a distinction between brightness of the red, yellow, and green objects to determine an object with the highest brightness. The object with the highest brightness may be assumed to be on, for example. In an instance in which multiple traffic signals are detected, a state of a traffic signal corresponding to a traffic lane of the vehicle may be determined. Additionally, in some instances, one or more color sensors may be used to

determine the state of a traffic signal. For example, using a gray (clear) sensor and a color-filtered sensor (e.g., a red filtered sensor) one can tell the difference between red and green objects. A red object will appear bright in a gray and red-filtered image while a green object will appear bright in a gray filtered image but much dimmer in the red-filtered image. It will be understood that the specific structure of red, yellow, and green lights is merely an example. Traffic signals may have varied and sometimes complex geometries and these additional geometries may also be detected. [0028] In an example in which a three-dimensional point cloud is determined, an object in the 3D point cloud having a shape that is expected for traffic signals may be detected. For example, 3D point cloud based object recognition systems may be used to identify features describing interest points within the 3D point cloud. The features may subsequently be compared to features expected for a 3D model of one or more types of traffic signals to recognize groups of interest points as objects within the 3D point cloud that are traffic signals. Given a recognized traffic signal, color information (e.g., an image) associated with the location of the 3D object may be analyzed to determine the state of the traffic signal.

[0029] Although the state of the traffic signal may be determined in some instances, there may be some degree of noise/uncertainty in the measurements and estimate of the state of the traffic signal. For example, a sensor used to determine the state of the traffic signal may include an amount of noise. Additionally, in some instances, a location and/or orientation of the traffic signal may change, impacting a clarity of the traffic signal in an image. As an example, a traffic signal may swing when suspended from a cable, or a traffic signal may be moved to a temporary location due to road construction. In another instance, the brightness or clarity of an image may be affected based on a position of the sun at various times of the day or night. In still another example, an image of a traffic signal may be noisy in various weather conditions such as fog, snow, rain, etc. Also, a tree or another vehicle may obstruct or partially obstruct a portion of a traffic signal in an image.

[0030] In some instances, information about the states of other traffic signals may be determined to aide in the determination of the state of the upcoming traffic signal or increase a confidence in an estimate of the state of the upcoming traffic signal. Although the method **100** is described with respect to an upcoming traffic signal, it is also contemplated that method **100** may be used to determine the state of any target traffic signal in a road network. For example, a navigation or route planning system may wish to know the state of any traffic signal in a network.

[0031] At block **104**, the method **100** includes determining, using the computing device, a state of one or more traffic signals other than the upcoming traffic signal. For instance, the one or more traffic signals may be other traffic signals in the environment. In some examples, the traffic signals may be traffic signals that are within a distance threshold of the upcoming threshold (e.g., within 2-3 blocks, within a mile, etc.) for which states have recently been observed.

[0032] As an example, the vehicle may obtain sensor data from sensors of the vehicle, and the state of the other traffic signals may be determined based on the sensor data. The vehicle may be able to determine the state of a cross-traffic traffic signal at an intersection in some cases. In other cases, the vehicle may be able to determine the state of a traffic signal which is the next traffic signal on a road (e.g., the traffic signal at the next intersection in the direction of travel) or recall the state of a traffic signal from a previous intersection from a memory.

[0033] As another example, the vehicle may receive information about the state of other traffic signals from one or more other vehicles. Other vehicles may observe a state of a traffic signal and send information to the vehicle directly (e.g., wirelessly), via a communication network, or via a server in a network. As still another example, fixed sensors in the environment may be configured to determine the state of other traffic signals and send the state of the other traffic signals to the vehicle, either directly or via any type of network. For instance, the fixed sensors may include imaging devices configured to observe the state of a traffic signal.

[0034] In other examples, the state of one or more traffic signals other than the traffic signal may

be received from transportation or road authorities. For instance, a transportation or road authority that manages a traffic control system for a road network may provide information about the states of one or more traffic signals that can be accessed in real-time. In some instances, because the information from the transportation or road authorities may be provided over a high latency channel, an estimate of the state of the traffic signal may account for the latency, and optionally make predictions based on the latency. For example, information indicating that a given traffic signal operating on a thirty second cycle changed to red four seconds ago may be received over a channel with a five to ten second latency. Accordingly, the estimate of the state of the traffic signal may be red. However, if the received information for the same channel indicated that the given traffic signal changed to red twenty-two seconds ago, the estimate of the state of the traffic signal may be green or red with associated probabilities based on average tendencies of the latency.

[0035] In another example, information received from transit authorities may provide an additional clue about the state of a traffic signal. For instance, information about the arrival or schedules of light rail vehicles, trains, cable-cars or other transit vehicles may be received. In cases in which transit vehicles intersect roads, the arrival of a transit vehicle at an intersection with a road may coincide with a state(s) of one or more traffic signals governing the flow of traffic on the road or adjacent roads.

[0036] In still other examples, one or more traffic signals may be configured to transmit their state and timing information via radio communication protocols. In a scenario in which the upcoming traffic signal is not configured to transmit state and timing information, information received from other traffic signals that do transmit state and timing information may be received.

[0037] At block **106**, the method **100** includes determining, using the computing device, an estimate of a state of the upcoming traffic signal based on a relationship between the state of the one or more traffic signals and the state of the upcoming traffic signal. In some cases, a conditional relationship between the state of the upcoming traffic signal and one or more other traffic signals may be known. As a simple example, it may be known that if the cross-traffic traffic signal at an intersection is green, the state of the upcoming traffic signal is red.

[0038] In another instance, the upcoming traffic signal may have a relationship with a traffic signal on an adjacent parallel road such that if the adjacent traffic signal is green, the upcoming traffic signal will be green, optionally with probability “P”. Similarly, a relationship between a past traffic signal on a road and the upcoming traffic signal may exist such that the upcoming traffic signal turns green “x” seconds after the previous traffic signal turns green. In some examples, the method **100** might also include determining an additional estimate of when the state of the upcoming traffic signal will change. For example, if another vehicle observed and recorded a time when the past traffic signal turned green, the computing device may be able to estimate that the upcoming traffic signal will turn green at a time that is “x” seconds from the recorded time.

[0039] By constructing a conditional map of traffic signals and their states based on observations of states of traffic signals or information about traffic signal patterns received from a transportation or road authority, and observing states of other traffic signals in real-time, the vehicle or another device may be provided with useful additional information to improve determination of the state of the upcoming traffic signal. There are many ways to create such a network map including collecting data from vehicles and post-processing the data, acquiring data from transportation/road authorities, and/or setting up static sensors in the environment to acquire data. In further examples, the state of an upcoming traffic signal might also be determined based on a conditional relationship between states of other traffic signals as well as one or more other factors such as time of day, day of the week, or traffic conditions. There are also many ways of taking the network information and extracting useful pieces of information for subsequent state estimation. Conditioning the state of each traffic signal on the state of all other traffic signals is one approach. Another approach might involve conditioning the state of each traffic signal based on a small fraction of other traffic signals (e.g., those in close proximity or within a distance threshold of a given upcoming traffic signal).

[0040] At block **108**, the method **100** includes the computing device sending the estimate of the state of the upcoming traffic signal to another device. As an example, the estimate of the state of the upcoming traffic signal may be sent to a controller that is configured to control the vehicle in an autonomous mode based on the estimate of the state of the upcoming traffic signal. In some examples, the vehicle may use the estimate of the state of the upcoming traffic signal as a check on a determined state of the upcoming traffic signal. As an example, if the estimate of the traffic signal is green, and the state of the traffic signal as determined by the vehicle is green, the vehicle may proceed through the traffic intersection and maintain its speed. Alternatively, if the state of the upcoming traffic signal as determined by the vehicle is green, and the estimate of the state of the upcoming traffic signal is red, the vehicle may proceed with caution, or seek additional information regarding the state of the traffic signal from any number of other sources.

[0041] For instance, the vehicle may gather information (e.g., determined in real-time or previously from the one or more sensors or additional sensors) to confirm the state of the traffic signal. The information may indicate behavior of other vehicles, such as vehicles traveling in the same or opposite direction, or cross-traffic. In another example, the vehicle may determine information such as changes in speed or acceleration, or distances to vehicles in front of and/or behind the vehicle. A vehicle in front of the vehicle that is braking may be indicative of a traffic signal that is yellow or red. The information may also indicate behavior of pedestrians in a crosswalk. For example, a pedestrian walking in a crosswalk that is perpendicular to a direction of travel of the vehicle may be indicative of a traffic signal that is red. In some instances, the vehicle may move to an adjusted position and determine additional information. For instance, the vehicle may move to an adjacent lane that is also a path in the same direction through the traffic intersection and determine information about a state of the traffic signal or a state of a different traffic signal. In other instances, the vehicle may gather information from pedestrian crosswalk signs which may display or flash a message (e.g., “walk” or “don't walk”) or include a countdown timer. Based on the orientation of and an observed state of the crosswalk sign and an intended route of the vehicle through an intersection, an additional inference regarding the state of the traffic intersection or traffic signal may be determined. For example, seeing a “walk” message in a direction that agrees with a direction of the vehicle may offer a clue, but not a certainty due to pedestrian's having the right-of-way in some instances, that a traffic signal is green.

[0042] In other examples, the vehicle may request information from a passenger or driver of the vehicle regarding the state of a traffic intersection or a traffic signal. Based on information received from the passenger or driver, the behavior of the vehicle may be modified. In other examples, the vehicle may tap into a network of the traffic signal or multiple traffic signals to determine information about the state of the traffic signal. For example, the vehicle may receive information about the state of a traffic signal from a wireless transmitter or transceiver located at the traffic signal, at a wireless tower, on a satellite, on another vehicle, etc. Thus, the example method **100** may enable controlling a vehicle based on an estimate of a state of an upcoming traffic signal that is determined based on states of other traffic signals.

[0043] As another example, the computing device may provide the estimate of the state of the upcoming traffic signal to a driver assistance system. For instance, the driver assistance system may include an intelligent speed adaptation system that is configured to adjust the speed of the vehicle based on an estimate of the state of the upcoming traffic signal and optionally an estimate of when the state of the upcoming traffic signal will change. In another instance, the driver assistance system may include a navigation system that is configured to plan a route for a driver of the vehicle based on an estimate of the state of the upcoming traffic signal. As an example, the navigation system may select a different route based on the estimate of the state of the upcoming traffic signal. In still another instance, the driver assistance system may be configured to alert the driver of the vehicle via an audible, visual, or other type of alert regarding the state of the upcoming traffic signal. Other example driver assistance systems are also contemplated.

[0044] One or more blocks **102-108** of the method **100** may be performed by a computing device in a server or by a computing device of the vehicle. For example, a computing device in a server may determine that the vehicle is approaching the upcoming traffic signal, determine the state of one or more traffic signals other than the upcoming traffic signal, and determine the estimate of the state of the upcoming traffic signal. In one instance, the computing device may access a conditional map having relationships between states of traffic signals, determine the state estimate based on the relationships, and provide the estimate to the computing device. Additionally, the computing device may provide instructions for controlling the vehicle in the autonomous mode to the vehicle.

[0045] A number of example scenarios and implementations of the method **100** are described below in connection with FIGS. **2A-4**. For purposes of illustration, multiple example implementations are described. It is to be understood, however, that the example implementations are illustrative only and are not meant to limiting. Other example implementations are possible as well.

[0046] FIGS. **2A-2C** are example conceptual illustrations of determining a state of a traffic signal. For example, FIGS. **2A-2C** illustrate a top view of a vehicle **200** and two perspective views from a position of the vehicle **200**. In some examples, the vehicle **200** may scan a target area to determine information associated with a state of a traffic signal. For example, information within a field of view **202** of the vehicle **200** may be obtained. In one instance, an imaging component and/or radar component of the vehicle may be used to obtain information within the field of view **202**. The imaging component and radar component may be mounted behind or adjacent to a rear-view mirror, such that in a scenario in which the vehicle **200** includes a driver and/or passenger(s), the imaging component or radar component minimally obstructs a field of view of the driver and/or passenger(s). In another scenario, the imaging component and/or the radar component may be mounted on the top of the vehicle **200**.

[0047] As shown in FIG. **2B**, in one example, the vehicle **200** may obtain an image of an area **204B**. The image may include 2D and/or 3D information. In some examples, the area **204B** may include one or more traffic signals and may optionally be predetermined based on known locations of traffic signals and a location and orientation of the vehicle **200**.

[0048] In some examples, the clarity of an image of the area **204B** may be affected by lighting conditions. As shown in FIG. **2B**, the position of the sun may be located behind a position of a traffic signal at some times of the day. This may lead to an uncertainty that is higher than uncertainties for other times of the day when the clarity of the image is not affected by the position of the sun.

[0049] As shown in FIG. **2C**, an image of an area **204C** may also be obstructed by physical objects. For example, a tree limb may obstruct a portion of a traffic signal in an image of the area **204C**. In one example, a 3D image of the area **204C** may reveal objects at multiple distances from the position of the vehicle in a region for which a traffic signal is expected. For example, a first distance may be determined to a point in the area **204C** that is closer to the vehicle than a group of points in the area **204C** that are in the shape of a portion of a traffic signal. This may be indicative of an object that is obstructing a view of the traffic signal. In some instances, a high uncertainty may be determined for instances in which an object is obstructing a view of the traffic signal.

[0050] In some example scenarios, if a view of a traffic signal is obstructed, or information associated with the state of the traffic signal is noisy, an estimate of the state of the traffic intersection may be determined based on states of other traffic signals. FIG. **3** is an example flow chart **300** for controlling a vehicle.

[0051] As shown in FIG. **3**, at block **302**, a determination may be made, by the vehicle and/or a server, whether the vehicle is approaching a traffic signal. For instance, the determination may be made based on a location of the vehicle in relation to a known map having locations of traffic signals. If the vehicle is not approaching a traffic signal, the determination may again be made at a later instance in time. If the vehicle is approaching the traffic signal, at block **304a**, a server may request information for states of other traffic signals. The states may include recently observed

states of other traffic signals (e.g., states observed within the past second, five seconds, etc.) that are provided to the server by vehicles, fixed sensors, transportation or road authorities, or other sources. Additionally, in parallel, the vehicle may attempt to detect the traffic signal at block **304b**. For instance, the vehicle may capture an image of a target area and process the image.

[0052] The server may also determine whether other states of traffic signals have recently been observed at block **306**. If information about the other states is not received, the vehicle may proceed to identify the state of the traffic signal, at block **308**, based on information from the sensors of the vehicle. Subsequently, based on the state of the traffic signal, the vehicle may be controlled in the autonomous mode at block **310**. If at block **306**, information about other states of traffic signals is received, the server may, at block **312**, determine an estimate of the state of the traffic signal that the vehicle is approaching. The estimate may be determined based on a relationship between the states of the other traffic signals and the state of the traffic signal. The estimate may be provided to the vehicle, such that at block **308**, the vehicle may identify the state of the traffic signal based on any detected information from the sensors of the vehicle as well as the estimate determined at block **312**.

[0053] FIG. **4** is an example conceptual illustration of determining states of traffic signals in a road network to estimate a state of an upcoming traffic signal. As shown in FIG. **4**, a primary vehicle **400** may be approaching an upcoming traffic signal **402**. To improve the estimate of a state of a traffic signal, information about observed states of any of secondary traffic signals **404a-e** in the environment may be utilized.

[0054] In one example, primary vehicle **400** may be able to observe the state of secondary traffic signal **404a** and/or **404b** while approaching upcoming traffic signal **402**. In an instance in which primary vehicle **400** observes the state of traffic signal **402a**, information about a relationship between the state of secondary traffic signal **404a** and the state of upcoming traffic signal **402** may be used to estimate the state of upcoming traffic signal **402**. For instance, secondary traffic signal **404a** and upcoming traffic signal **402** may turn green at the same time. In an instance in which primary vehicle **400** observes the state of secondary traffic signal **404b**, an estimate of the state of the upcoming traffic signal **402** may be determined based on a relationship between the states of upcoming traffic signal **402** and secondary traffic signal **404b**. For instance, the upcoming traffic signal **402** may be red while the secondary traffic signal **404b** is green or yellow, and may turn green one second after secondary traffic signal **404b** turns red.

[0055] In another example, the primary vehicle may have recently observed the state of secondary traffic signal **404c** and rely on a relationship between the states of upcoming traffic signal **402** and the secondary traffic signal **404c** to determine an estimate of the state of the upcoming traffic signal **402**. For instance, the relationship between the states of secondary traffic signal **404c** and upcoming traffic signal **402** may be that the upcoming traffic signal **402** turns green four seconds after the secondary traffic signal **404c** turns green. If the primary vehicle observes and records the time when the secondary traffic signal **404c** turns green, the primary vehicle **400** may be able to estimate that the upcoming traffic signal is green at an instance in time that is five seconds after the time when the secondary traffic signal **404c** turned green.

[0056] Additionally, secondary vehicles **406a** and **406b** may observe the state of secondary traffic signals **404b** and/or **404d** respectively, and transmit the state information to the primary vehicle **400** or a server in a network via the communication tower **408**. In still another example, a fixed sensor **410** may observe the states of secondary traffic signals **404a** and/or **404e** and transmit the information to the primary vehicle **400** or a server in a network via satellite **412**. Thus, the primary vehicle **400** and/or a server may receive information about observed states of other traffic signals in a network that may be used to improve an estimate of a state of the upcoming traffic signal **402**.

[0057] Systems in which example embodiments of the above example methods may be implemented will now be described in greater detail. In general, an example system may be implemented in or may take the form of a vehicle. The vehicle may take a number of forms,

including, for example, automobiles, cars, trucks, motorcycles, buses, boats, airplanes, helicopters, lawn mowers, earth movers, snowmobiles, recreational vehicles, amusement park vehicles, farm equipment, construction equipment, trams, golf carts, trains, and trolleys. Other vehicles are possible as well.

[0058] Further, another example system may take the form of non-transitory computer-readable medium, which has program instructions stored thereon that are executable by at least one processor to provide the functionality described herein. An example system may also take the form of a vehicle or a subsystem of a vehicle that includes such a non-transitory computer-readable medium having such program instructions stored thereon.

[0059] FIG. 5 illustrates an example vehicle **500**, in accordance with an embodiment. In particular, FIG. 5 shows a Right Side View, Front View, Back View, and Top View of the vehicle **500**. Although vehicle **500** is illustrated in FIG. 5 as a car, other embodiments are possible. For instance, the vehicle **500** could represent a truck, a van, a semi-trailer truck, a motorcycle, a golf cart, an off-road vehicle, or a farm vehicle, among other examples. As shown, the vehicle **500** includes a first sensor unit **502**, a second sensor unit **504**, a third sensor unit **506**, a wireless communication system **508**, and a camera **510**.

[0060] Each of the first, second, and third sensor units **502-506** may include any combination of global positioning system sensors, inertial measurement units, radio detection and ranging (RADAR) units, laser rangefinders, light detection and ranging (LIDAR) units, cameras, and acoustic sensors. Other types of sensors are possible as well.

[0061] While the first, second, and third sensor units **502** are shown to be mounted in particular locations on the vehicle **500**, in some embodiments the sensor unit **502** may be mounted elsewhere on the vehicle **500**, either inside or outside the vehicle **500**. Further, while only three sensor units are shown, in some embodiments more or fewer sensor units may be included in the vehicle **500**.

[0062] In some embodiments, one or more of the first, second, and third sensor units **502-506** may include one or more movable mounts on which the sensors may be movably mounted. The movable mount may include, for example, a rotating platform. Sensors mounted on the rotating platform could be rotated so that the sensors may obtain information from each direction around the vehicle **500**. Alternatively or additionally, the movable mount may include a tilting platform. Sensors mounted on the tilting platform could be tilted within a particular range of angles and/or azimuths so that the sensors may obtain information from a variety of angles. The movable mount may take other forms as well.

[0063] Further, in some embodiments, one or more of the first, second, and third sensor units **502-506** may include one or more actuators configured to adjust the position and/or orientation of sensors in the sensor unit by moving the sensors and/or movable mounts. Example actuators include motors, pneumatic actuators, hydraulic pistons, relays, solenoids, and piezoelectric actuators. Other actuators are possible as well.

[0064] The wireless communication system **508** may be any system configured to wirelessly couple to one or more other vehicles, sensors, or other entities, either directly or via a communication network. To this end, the wireless communication system **508** may include an antenna and a chipset for communicating with the other vehicles, sensors, or other entities either directly or via a communication network. The chipset or wireless communication system **508** in general may be arranged to communicate according to one or more other types of wireless communication (e.g., protocols) such as Bluetooth, communication protocols described in IEEE 802.11 (including any IEEE 802.11 revisions), cellular technology (such as GSM, CDMA, UMTS, EV-DO, WiMAX, or LTE), Zigbee, dedicated short range communications (DSRC), and radio frequency identification (RFID) communications, among other possibilities. The wireless communication system **508** may take other forms as well.

[0065] While the wireless communication system **508** is shown positioned on a roof of the vehicle **500**, in other embodiments the wireless communication system **508** could be located, fully or in

part, elsewhere.

[0066] The camera **510** may be any camera (e.g., a still camera, a video camera, etc.) configured to capture images of the environment in which the vehicle **500** is located. To this end, the camera **510** may be configured to detect visible light, or may be configured to detect light from other portions of the spectrum, such as infrared or ultraviolet light. Other types of cameras are possible as well. The camera **510** may be a two-dimensional detector, or may have a three-dimensional spatial range. In some embodiments, the camera **510** may be, for example, a range detector configured to generate a two-dimensional image indicating a distance from the camera **510** to a number of points in the environment. To this end, the camera **510** may use one or more range detecting techniques. For example, the camera **510** may use a structured light technique in which the vehicle **500** illuminates an object in the environment with a predetermined light pattern, such as a grid or checkerboard pattern and uses the camera **510** to detect a reflection of the predetermined light pattern off the object. Based on distortions in the reflected light pattern, the vehicle **500** may determine the distance to the points on the object. The predetermined light pattern may comprise infrared light, or light of another wavelength. As another example, the camera **510** may use a laser scanning technique in which the vehicle **500** emits a laser and scans across a number of points on an object in the environment. While scanning the object, the vehicle **500** uses the camera **510** to detect a reflection of the laser off the object for each point. Based on a length of time it takes the laser to reflect off the object at each point, the vehicle **500** may determine the distance to the points on the object. As yet another example, the camera **510** may use a time-of-flight technique in which the vehicle **500** emits a light pulse and uses the camera **510** to detect a reflection of the light pulse off an object at a number of points on the object. In particular, the camera **510** may include a number of pixels, and each pixel may detect the reflection of the light pulse from a point on the object. Based on a length of time it takes the light pulse to reflect off the object at each point, the vehicle **500** may determine the distance to the points on the object. The light pulse may be a laser pulse. Other range detecting techniques are possible as well, including stereo triangulation, sheet-of-light triangulation, interferometry, and coded aperture techniques, among others. The camera **510** may take other forms as well.

[0067] In some embodiments, the camera **510** may include a movable mount and/or an actuator, as described above, that are configured to adjust the position and/or orientation of the camera **510** by moving the camera **510** and/or the movable mount.

[0068] While the camera **510** is shown to be mounted inside a front windshield of the vehicle **500**, in other embodiments the camera **510** may be mounted elsewhere on the vehicle **500**, either inside or outside the vehicle **500**.

[0069] The vehicle **500** may include one or more other components in addition to or instead of those shown.

[0070] FIG. **6** is a simplified block diagram of an example vehicle **600**, in accordance with an embodiment. The vehicle **600** may, for example, be similar to the vehicle **500** described above in connection with FIG. **5**. The vehicle **600** may take other forms as well.

[0071] As shown, the vehicle **600** includes a propulsion system **602**, a sensor system **604**, a control system **606**, peripherals **608**, and a computer system **610** including a processor **612**, data storage **614**, and instructions **616**. In other embodiments, the vehicle **600** may include more, fewer, or different systems, and each system may include more, fewer, or different components. Additionally, the systems and components shown may be combined or divided in any number of ways.

[0072] The propulsion system **602** may be configured to provide powered motion for the vehicle **600**. As shown, the propulsion system **602** includes an engine/motor **618**, an energy source **620**, a transmission **622**, and wheels/tires **624**.

[0073] The engine/motor **618** may be or include any combination of an internal combustion engine, an electric motor, a steam engine, and a Stirling engine. Other motors and engines are possible as well. In some embodiments, the propulsion system **602** could include multiple types of engines

and/or motors. For instance, a gas-electric hybrid car could include a gasoline engine and an electric motor. Other examples are possible.

[0074] The energy source **620** may be a source of energy that powers the engine/motor **618** in full or in part. That is, the engine/motor **618** may be configured to convert the energy source **620** into mechanical energy. Examples of energy sources **620** include gasoline, diesel, propane, other compressed gas-based fuels, ethanol, solar panels, batteries, and other sources of electrical power. The energy source(s) **620** could additionally or alternatively include any combination of fuel tanks, batteries, capacitors, and/or flywheels. In some embodiments, the energy source **620** may provide energy for other systems of the vehicle **600** as well.

[0075] The transmission **622** may be configured to transmit mechanical power from the engine/motor **618** to the wheels/tires **624**. To this end, the transmission **622** may include a gearbox, clutch, differential, drive shafts, and/or other elements. In embodiments where the transmission **622** includes drive shafts, the drive shafts could include one or more axles that are configured to be coupled to the wheels/tires **624**.

[0076] The wheels/tires **624** of vehicle **600** could be configured in various formats, including a unicycle, bicycle/motorcycle, tricycle, or car/truck four-wheel format. Other wheel/tire formats are possible as well, such as those including six or more wheels. In any case, the wheels/tires **624** of vehicle **624** may be configured to rotate differentially with respect to other wheels/tires **624**. In some embodiments, the wheels/tires **624** may include at least one wheel that is fixedly attached to the transmission **622** and at least one tire coupled to a rim of the wheel that could make contact with the driving surface. The wheels/tires **624** may include any combination of metal and rubber, or combination of other materials.

[0077] The propulsion system **602** may additionally or alternatively include components other than those shown.

[0078] The sensor system **604** may include a number of sensors configured to sense information about an environment in which the vehicle **600** is located, as well as one or more actuators **636** configured to modify a position and/or orientation of the sensors. As shown, the sensors of the sensor system include a Global Positioning System (GPS) **626**, an inertial measurement unit (IMU) **628**, a RADAR unit **630**, a laser rangefinder and/or LIDAR unit **632**, and a camera **634**. The sensor system **604** may include additional sensors as well, including, for example, sensors that monitor internal systems of the vehicle **600** (e.g., an O.sub.2 monitor, a fuel gauge, an engine oil temperature, etc.). Other sensors are possible as well.

[0079] The GPS **626** may be any sensor configured to estimate a geographic location of the vehicle **600**. To this end, the GPS **626** may include a transceiver configured to estimate a position of the vehicle **600** with respect to the Earth. The GPS **626** may take other forms as well.

[0080] The IMU **628** may be any combination of sensors configured to sense position and orientation changes of the vehicle **600** based on inertial acceleration. In some embodiments, the combination of sensors may include, for example, accelerometers and gyroscopes. Other combinations of sensors are possible as well.

[0081] The RADAR **630** unit may be any sensor configured to sense objects in the environment in which the vehicle **600** is located using radio signals. In some embodiments, in addition to sensing the objects, the RADAR unit **630** may additionally be configured to sense the speed and/or heading of the objects.

[0082] Similarly, the laser rangefinder or LIDAR unit **632** may be any sensor configured to sense objects in the environment in which the vehicle **600** is located using lasers. In particular, the laser rangefinder or LIDAR unit **632** may include a laser source and/or laser scanner configured to emit a laser and a detector configured to detect reflections of the laser. The laser rangefinder or LIDAR **632** may be configured to operate in a coherent (e.g., using heterodyne detection) or an incoherent detection mode.

[0083] The camera **634** may be any camera (e.g., a still camera, a video camera, etc.) configured to

capture images of the environment in which the vehicle **600** is located. To this end, the camera may take any of the forms described above.

[0084] The sensor system **604** may additionally or alternatively include components other than those shown.

[0085] The control system **606** may be configured to control operation of the vehicle **600** and its components. To this end, the control system **606** may include a steering unit **638**, a throttle **640**, a brake unit **642**, a sensor fusion algorithm **644**, a computer vision system **646**, a navigation or pathing system **648**, and an obstacle avoidance system **650**.

[0086] The steering unit **638** may be any combination of mechanisms configured to adjust the heading of vehicle **600**.

[0087] The throttle **640** may be any combination of mechanisms configured to control the operating speed of the engine/motor **618** and, in turn, the speed of the vehicle **600**.

[0088] The brake unit **642** may be any combination of mechanisms configured to decelerate the vehicle **600**. For example, the brake unit **642** may use friction to slow the wheels/tires **624**. As another example, the brake unit **642** may convert the kinetic energy of the wheels/tires **624** to electric current. The brake unit **642** may take other forms as well.

[0089] The sensor fusion algorithm **644** may be an algorithm (or a computer program product storing an algorithm) configured to accept data from the sensor system **604** as an input. The data may include, for example, data representing information sensed at the sensors of the sensor system **604**. The sensor fusion algorithm **644** may include, for example, a Kalman filter, a Bayesian network, or another algorithm. The sensor fusion algorithm **644** may further be configured to provide various assessments based on the data from the sensor system **604**, including, for example, evaluations of individual objects and/or features in the environment in which the vehicle **600** is located, evaluations of particular situations, and/or evaluations of possible impacts based on particular situations. Other assessments are possible as well.

[0090] The computer vision system **646** may be any system configured to process and analyze images captured by the camera **634** in order to identify objects and/or features in the environment in which the vehicle **600** is located, including, for example, traffic signals and obstacles. To this end, the computer vision system **646** may use an object recognition algorithm, a Structure from Motion (SFM) algorithm, video tracking, or other computer vision techniques. In some embodiments, the computer vision system **646** may additionally be configured to map the environment, track objects, estimate the speed of objects, etc.

[0091] The navigation and pathing system **648** may be any system configured to determine a driving path for the vehicle **600**. The navigation and pathing system **648** may additionally be configured to update the driving path dynamically while the vehicle **600** is in operation. In some embodiments, the navigation and pathing system **648** may be configured to incorporate data from the sensor fusion algorithm **644**, the GPS **626**, and one or more predetermined maps so as to determine the driving path for vehicle **100**.

[0092] The obstacle avoidance system **650** may be any system configured to identify, evaluate, and avoid or otherwise negotiate obstacles in the environment in which the vehicle **600** is located.

[0093] The control system **606** may additionally or alternatively include components other than those shown.

[0094] Peripherals **608** may be configured to allow the vehicle **600** to interact with external sensors, other vehicles, and/or a user. To this end, the peripherals **608** may include, for example, a wireless communication system **652**, a touchscreen **654**, a microphone **656**, and/or a speaker **658**.

[0095] The wireless communication system **652** may take any of the forms described above.

[0096] The touchscreen **654** may be used by a user to input commands to the vehicle **600**. To this end, the touchscreen **654** may be configured to sense at least one of a position and a movement of a user's finger via capacitive sensing, resistance sensing, or a surface acoustic wave process, among other possibilities. The touchscreen **654** may be capable of sensing finger movement in a direction

parallel or planar to the touchscreen surface, in a direction normal to the touchscreen surface, or both, and may also be capable of sensing a level of pressure applied to the touchscreen surface. The touchscreen **654** may be formed of one or more translucent or transparent insulating layers and one or more translucent or transparent conducting layers. The touchscreen **654** may take other forms as well.

[0097] The microphone **656** may be configured to receive audio (e.g., a voice command or other audio input) from a user of the vehicle **600**. Similarly, the speakers **658** may be configured to output audio to the user of the vehicle **600**.

[0098] The peripherals **608** may additionally or alternatively include components other than those shown.

[0099] The computer system **610** may be configured to transmit data to and receive data from one or more of the propulsion system **602**, the sensor system **604**, the control system **606**, and the peripherals **608**. To this end, the computer system **610** may be communicatively linked to one or more of the propulsion system **602**, the sensor system **604**, the control system **606**, and the peripherals **608** by a system bus, network, and/or other connection mechanism (not shown).

[0100] The computer system **610** may be further configured to interact with and control one or more components of the propulsion system **602**, the sensor system **604**, the control system **606**, and/or the peripherals **608**. For example, the computer system **610** may be configured to control operation of the transmission **622** to improve fuel efficiency. As another example, the computer system **610** may be configured to cause the camera **634** to capture images of the environment. As yet another example, the computer system **610** may be configured to store and execute instructions corresponding to the sensor fusion algorithm **644**. As still another example, the computer system **610** may be configured to store and execute instructions for displaying a display on the touchscreen **654**. Other examples are possible as well.

[0101] As shown, the computer system **610** includes the processor **612** and data storage **614**. The processor **612** may comprise one or more general-purpose processors and/or one or more special-purpose processors. To the extent the processor **612** includes more than one processor, such processors could work separately or in combination. Data storage **614**, in turn, may comprise one or more volatile and/or one or more non-volatile storage components, such as optical, magnetic, and/or organic storage, and data storage **614** may be integrated in whole or in part with the processor **612**.

[0102] In some embodiments, data storage **614** may contain instructions **616** (e.g., program logic) executable by the processor **612** to execute various vehicle functions, including those described above in connection with FIG. 1. Further, data storage **614** may contain constraints **660** for the vehicle **600**, which may take any of the forms described above. Data storage **614** may contain additional instructions as well, including instructions to transmit data to, receive data from, interact with, and/or control one or more of the propulsion system **602**, the sensor system **604**, the control system **606**, and the peripherals **608**.

[0103] The computer system **602** may additionally or alternatively include components other than those shown.

[0104] As shown, the vehicle **600** further includes a power supply **610**, which may be configured to provide power to some or all of the components of the vehicle **600**. To this end, the power supply **610** may include, for example, a rechargeable lithium-ion or lead-acid battery. In some embodiments, one or more banks of batteries could be configured to provide electrical power. Other power supply materials and configurations are possible as well. In some embodiments, the power supply **610** and energy source **620** may be implemented together, as in some all-electric cars.

[0105] In some embodiments, one or more of the propulsion system **602**, the sensor system **604**, the control system **606**, and the peripherals **608** could be configured to work in an interconnected fashion with other components within and/or outside their respective systems.

[0106] Further, the vehicle **600** may include one or more elements in addition to or instead of those

shown. For example, the vehicle **600** may include one or more additional interfaces and/or power supplies. Other additional components are possible as well. In such embodiments, data storage **614** may further include instructions executable by the processor **612** to control and/or communicate with the additional components.

[0107] Still further, while each of the components and systems are shown to be integrated in the vehicle **600**, in some embodiments, one or more components or systems may be removably mounted on or otherwise connected (mechanically or electrically) to the vehicle **600** using wired or wireless connections.

[0108] The vehicle **600** may take other forms as well.

[0109] In some embodiments, the disclosed methods may be implemented as computer program instructions encoded on a non-transitory computer-readable storage media in a machine-readable format, or on other non-transitory media or articles of manufacture. FIG. 7 is a schematic illustrating a conceptual partial view of an example computer program product **700** that includes a computer program for executing a computer process on a computing device, arranged according to at least some embodiments presented herein.

[0110] In one embodiment, the example computer program product **700** is provided using a signal bearing medium **702**. The signal bearing medium **702** may include one or more programming instructions **704** that, when executed by one or more processors, may provide functionality or portions of the functionality described above with respect to FIGS. 1-6.

[0111] In some embodiments, the signal bearing medium **702** may encompass a computer-readable medium **706**, such as, but not limited to, a hard disk drive, a Compact Disc (CD), a Digital Video Disk (DVD), a digital tape, memory, etc. Further, in some embodiments the signal bearing medium **702** may encompass a computer recordable medium **707**, such as, but not limited to, memory, read/write (R/W) CDs, R/W DVDs, etc. Still further, in some embodiments the signal bearing medium **702** may encompass a communications medium **710**, such as, but not limited to, a digital and/or an analog communication medium (e.g., a fiber optic cable, a waveguide, a wired communications link, a wireless communication link, etc.). Thus, for example, the signal bearing medium **702** may be conveyed by a wireless form of the communications medium **710**.

[0112] The one or more programming instructions **704** may be, for example, computer executable and/or logic implemented instructions. In some examples, a computing system such as the computing system **610** of FIG. 6 may be configured to provide various operations, functions, or actions in response to the programming instructions **704** being conveyed to the computing system **610** by one or more of the computer readable medium **706**, the computer recordable medium **707**, and/or the communications medium **710**.

[0113] It should be understood that arrangements described herein are for purposes of example only. As such, those skilled in the art will appreciate that other arrangements and other elements (e.g. machines, interfaces, functions, orders, and groupings of functions, etc.) can be used instead, and some elements may be omitted altogether according to the desired results. Further, many of the elements that are described are functional entities that may be implemented as discrete or distributed components or in conjunction with other components, in any suitable combination and location.

[0114] While various aspects and embodiments have been disclosed herein, other aspects and embodiments will be apparent to those skilled in the art. The various aspects and embodiments disclosed herein are for purposes of illustration and are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A method comprising: receiving, at a computing device coupled to a vehicle, sensor data representing an environment of the vehicle as the vehicle navigates a path; determining, based on

the sensor data, the vehicle is approaching an intersection that includes a traffic signal; determining, by the computing device, an estimate of when the traffic signal will change from the current state to a different state; and causing an interface of the vehicle to provide a visual alert conveying the estimate of when the traffic signal will change from the current state to the different state.

2. The method of claim 1, wherein the sensor data includes at least one of camera data, lidar data, or radar data.
3. The method of claim 1, further comprising: receiving the estimate of when the traffic signal will change from a traffic management system or a connected infrastructure device;
4. The method of claim 3, wherein the connected infrastructure device is the traffic signal.
5. The method of claim 1, further comprising: determining a relationship between states of the traffic signal and a second traffic signal; and determining the estimate of when the traffic signal will change based on the determined relationship.
6. The method of claim 1, further comprising determining a confidence level for the estimate of when the traffic signal will change based on additional sensor data.
7. The method of claim 1, wherein the visual alert is displayed via a driver assistance system of the vehicle.
8. The method of claim 1, further comprising: providing an audible alert in addition to the visual alert.
9. The method of claim 1, further comprising: receiving information representing the current state of the traffic signal and a duration for which the traffic signal has been in the current state.
10. The method of claim 1, further comprising: updating the estimate of when the traffic signal will change based on real-time traffic conditions.
11. The method of claim 1, further comprising: transmitting the estimate of when the traffic signal will change to one or more vehicles using vehicle-to-vehicle communication.
12. The method of claim 1, wherein determining the estimate of when the traffic signal will change from the current state to the different state comprises: determining the estimate using information received via radio communication.
13. The method of claim 1, further comprising: receiving timing information that indicates a given operating cycle of the traffic signal and an amount of time since the traffic signal changed to the current state; and determining the estimate of when the traffic signal will change from the current state to the different state based on the timing information.
14. The method of claim 1, further comprising: based on the sensor data, detecting the traffic signal; determining a predicted color of the traffic signal above a confidence level; and based on determining the predicted color of the traffic signal above the confidence level, determining the estimate of when the traffic signal will change from the current state to the different state.
15. A system comprising: a sensor coupled to a vehicle; and a computing device configured to: receive sensor data representing an environment of the vehicle as the vehicle navigates a path; determine, based on the sensor data, the vehicle is approaching an intersection that includes a traffic signal; determine an estimate of when the traffic signal will change from the current state to a different state; and cause an interface of the vehicle to provide a visual alert conveying the estimate of when the traffic signal will change from the current state to the different state.
16. The system of claim 15, wherein the computing device is further configured to: receive information conveying the estimate of when the traffic signal will change from an external source.
17. The system of claim 16, wherein the external source corresponds to a second vehicle or the traffic signal.
18. The system of claim 15, further comprising: determine a control strategy for the vehicle based on the estimate of when the traffic signal will change from the current state to the different state; and cause the vehicle to autonomously navigate based on the control strategy.
19. The system of claim 15, wherein the sensor comprises one or more cameras and one or more

lidar units.

20. A non-transitory computer-readable medium having stored therein instructions executable by a computing device to cause the computing device to perform functions comprising: receiving sensor data representing an environment of the vehicle as the vehicle navigates a path; determining, based on the sensor data, the vehicle is approaching an intersection that includes a traffic signal; determining an estimate of when the traffic signal will change from the current state to a different state; and causing an interface of the vehicle to provide a visual alert conveying the estimate of when the traffic signal will change from the current state to the different state.
